THE CONE

SOLID MENSURATION

prepared by:

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

70%

LONG QUIZ

30%

SEATWORK

0

ABSENT

3

45-min LATE

7 max

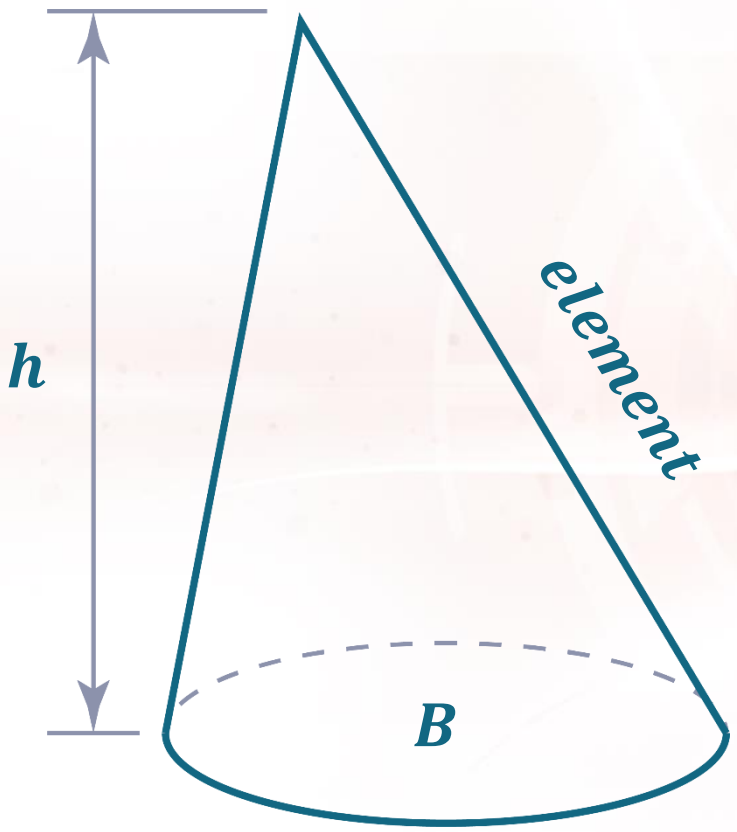
15-min LATE

TOPIC OUTLINE

Properties of a Cone

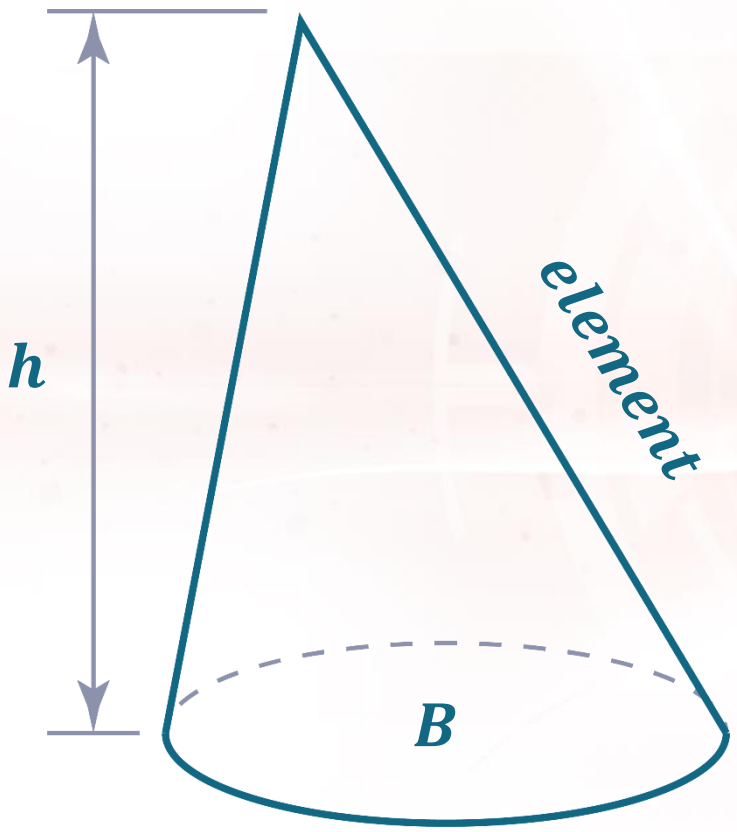
- Lateral Surface Area
- Total Surface Area
- Volume

PROPERTIES OF A CONE



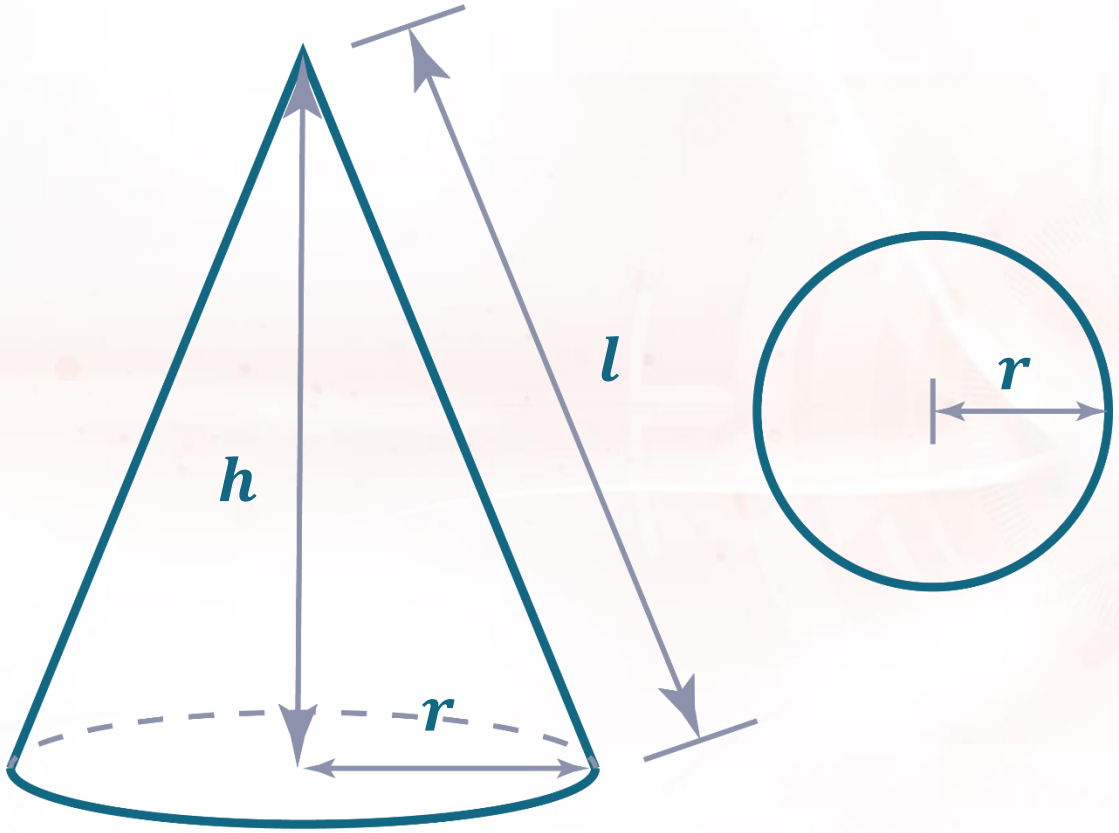
A **cone** is the solid bounded by a conical surface (**lateral surface**) whose directrix is a **closed curve**, and a plane (**base**) which cuts all the elements.

PROPERTIES



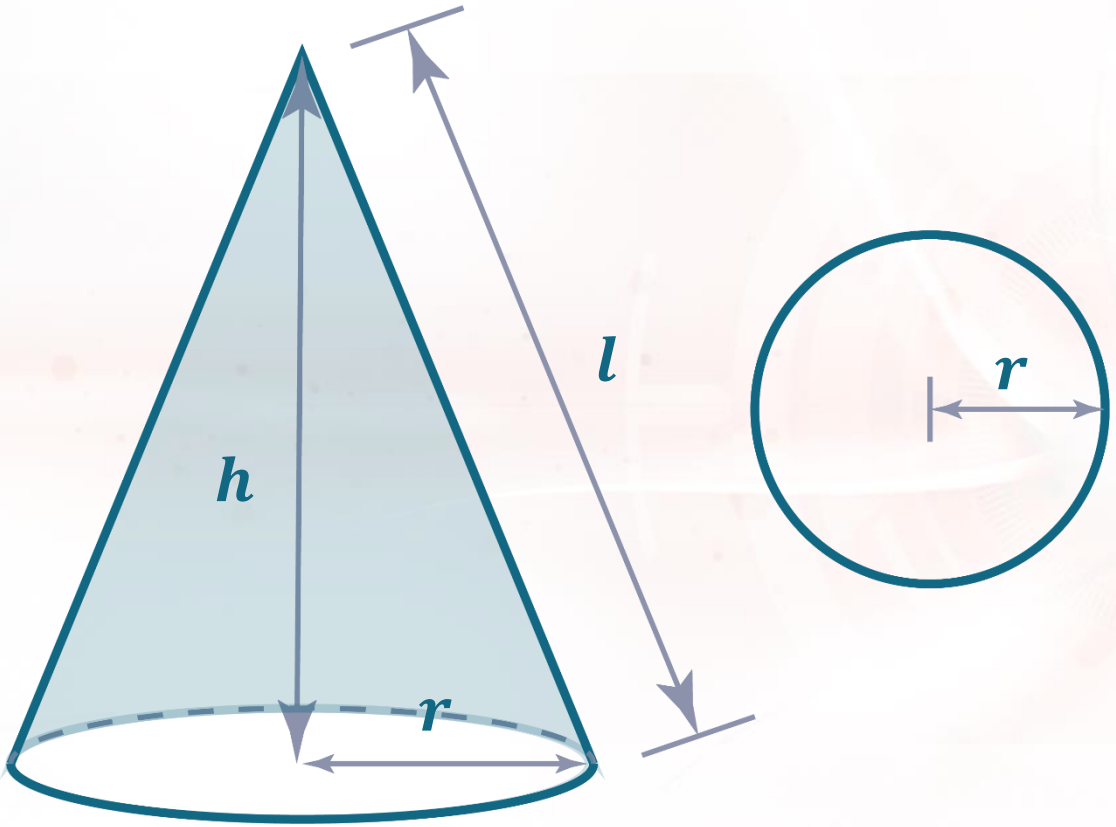
1. The **altitude, *h*** of a cone is the perpendicular distance from the vertex to the plane of the base.
2. Every section of a cone made by a plane passing through its vertex and containing two points of the base is a **triangle**.
3. A right section of a cone is a section perpendicular to its axis and cutting all the ***elements***.

AREA OF A RIGHT CIRCULAR CONE



A right circular cone is a cone whose base is a **circle** and whose axis is **perpendicular** to its base.

LATERAL SURFACE AREA



The Lateral Surface Area, **S**

$$S = eP$$

$$S = \pi r l$$

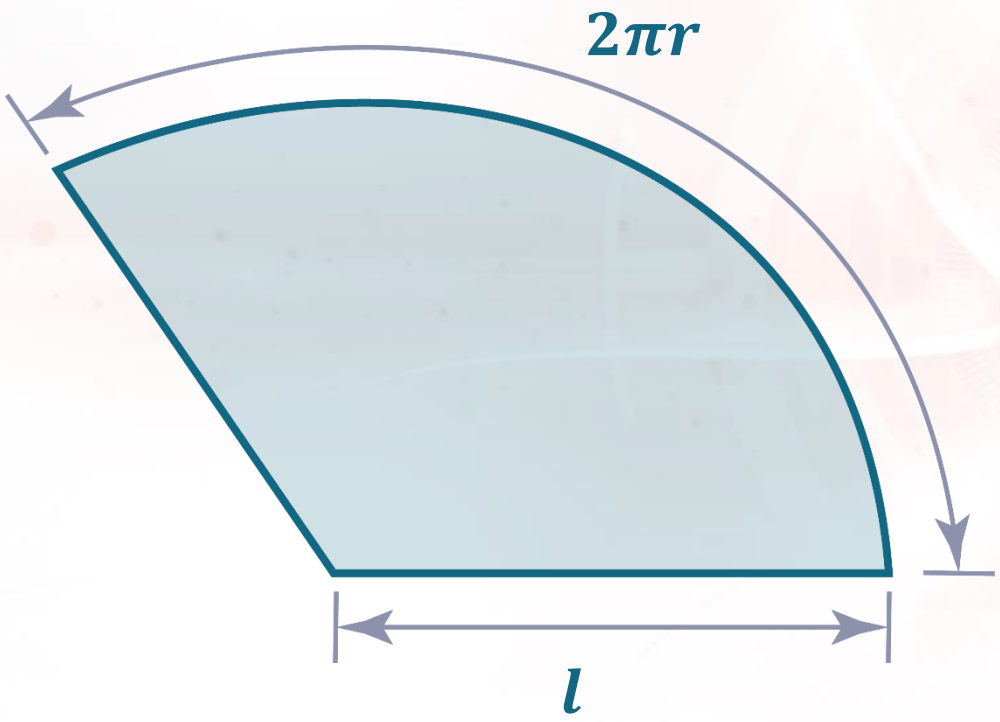
$$S = \pi r \sqrt{h^2 + r^2}$$

where,

r = radius of the base

l = slant height

h = altitude



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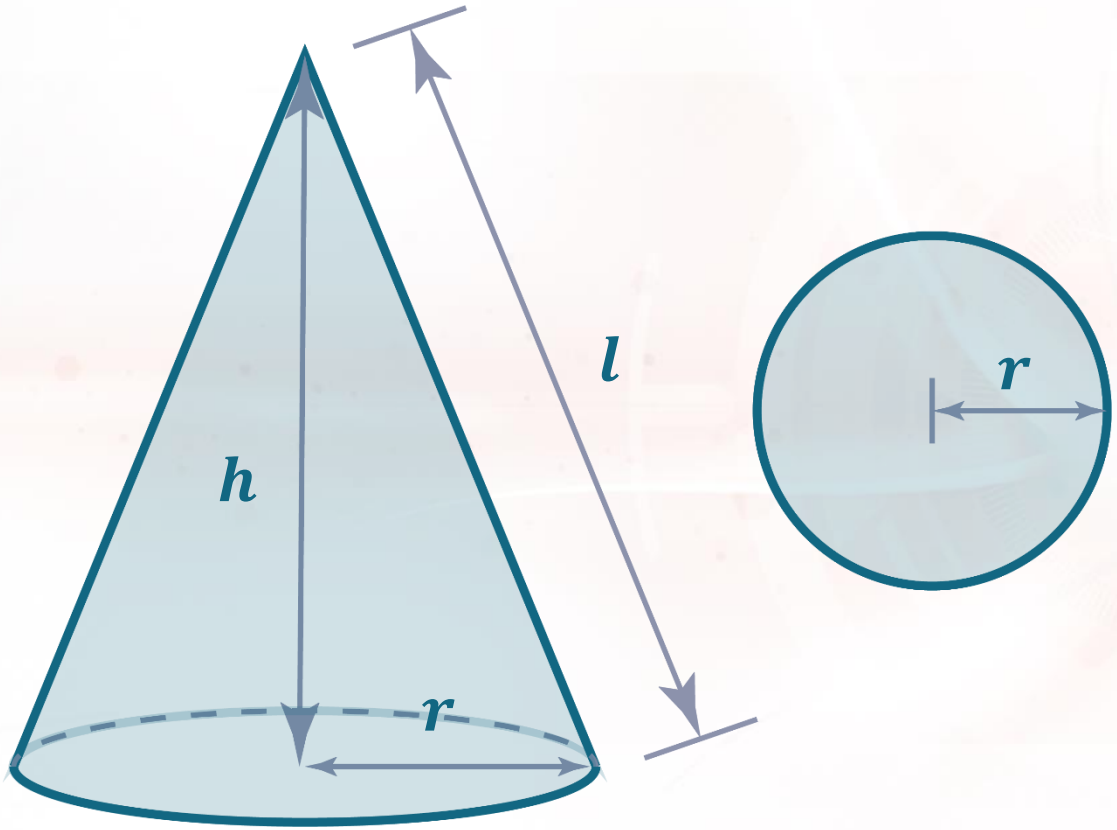
TOTAL SURFACE AREA



The Total Surface Area, **S**

$$T = S + B$$

$$T = \pi r l + \pi r^2$$

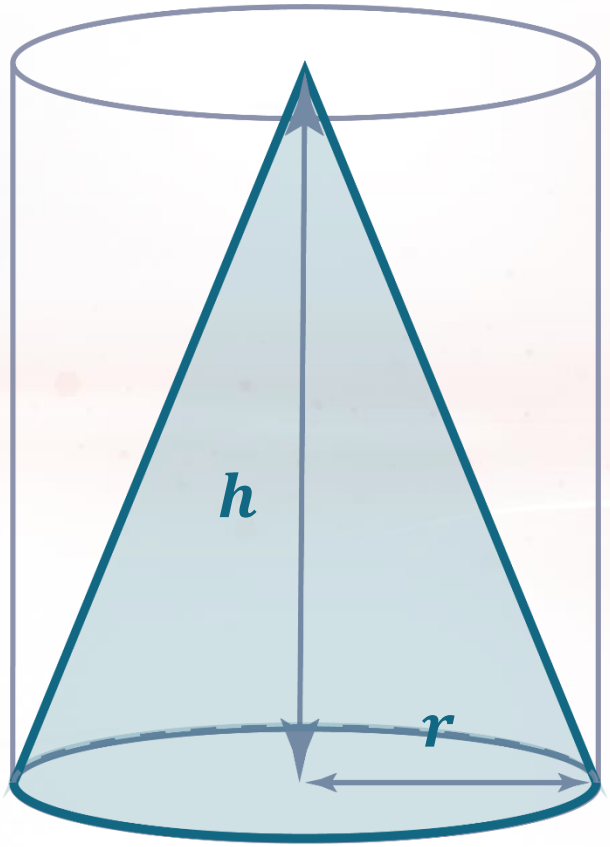
where,

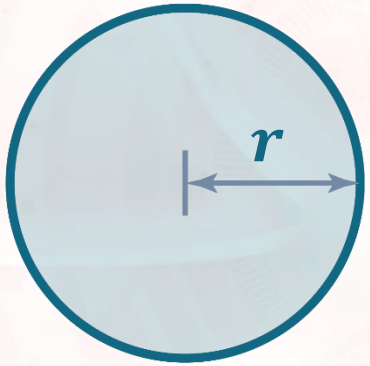
r = radius of the base

l = slant height

h = altitude

How many **square yards** of canvas will be required to make a **conical tent 15** ft. **high** and **18** ft. in **diameter**, if **10 percent** of the material is **wasted**?





The Volume, **V**

$$V = \frac{1}{3}Bh$$

$$V = \frac{1}{3}\pi r^2 h$$

where,

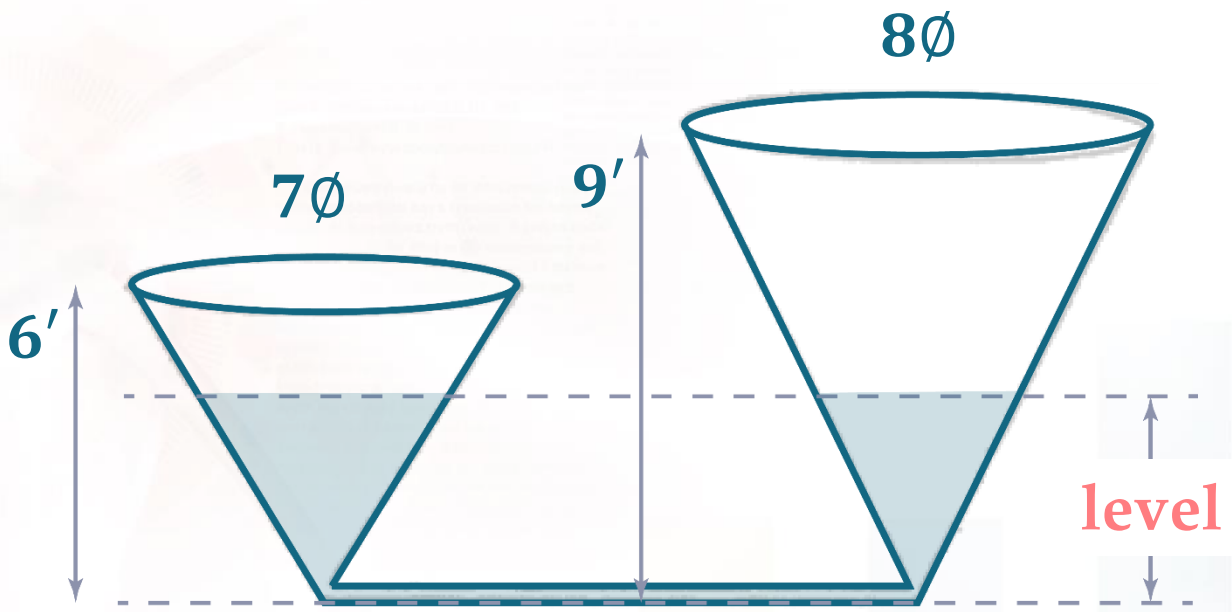
r = radius of the base

h = altitude

A pile of sand is in the form of a **right circular cone** of **altitude 7** ft. and **slant height 25** ft. What is the **weight** of the sand, if the sand weighs **107.5 lb. per cu. ft.**?

SEATWORK

1. A **circular** piece of filter paper **R** inches in radius is **folded twice**: first on a **diameter** and then on the **radius** perpendicular to the crease. It is then opened up so as to form a **right circular cone**. Find in terms of **R** the **volume** of liquid it will hold.

2. A stalactite which may be seen hanging in the Endless Caverns in Virginia is approximately in the form of a **cone 4** ft. **long** with a **base 18** in. in **diameter**. The calcium carbonate of which it is formed weighs **131 lb. per cu. ft**. Find the **weight** of the stalactite.

3. The two vertical **conical tanks** (both inverted) have their vertices connected by a short horizontal pipe. **One tank**, initially full of water, has an **altitude** of **6** ft. and a **diameter** of base **7** ft. The **other tank**, initially empty, has an **altitude** of **9** ft. and a **diameter** of base **8** ft. If the water is allowed to flow through the connecting pipe, find the **level** to which the **water** will ultimately rise in the empty tank. (Neglect the water in the pipe.)